Introduction to Software Engineering

Project Proposal & Plan

CourtMate | T05 Nexora | PA1 · Oct 2026

Software Engineering Department
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Project Proposal & Plan

Objectives

This proposal updates CourtMate's PA0 booking baseline with two team-selected enhancement ideas: Open Play Rooms (a deliberately small Must-have workflow) and Last-Minute Deals (a Should-have stretch feature). It sets out five core capability groups, the planned architecture, WBS/Gantt/CPM, estimates, risks, and teamwork requirements for PA1 review.

The baseline PA5 commitment is the booking system plus minimal booked-court room coordination; owner-managed Last-Minute Deals and profile editing are optional. PA2 will specify detailed rules and carry forward any approved changes.

1 Member Contribution Assessment

The four members and their GitHub contribution evidence are listed below. Authorship of a pull request and completion of an independent review are distinct evidence.

Student ID	Full name	PA1 role / responsibility	Share (%)	Evidence (GitHub)
24127174	Nguyễn Đức Hưng	Team Lead / Project Planning; Collaboration	25%	Issues #47/#49; PRs #54/#55; draft integration and documentation review.
24127519	Bùi Phạm Quang	Business Analyst / Vision; peer reviewer	25%	Issue #45; PR #52 (merged); reviews to be evidenced separately.
24127040	Dương Trung Hiếu	Designer / Preliminary Architecture	25%	Issue #46; PR #53 (merged); reviews to be evidenced separately.
24127139	Lê Thành Trung	Software Tester / Risk Management	25%	Issue #48; PR #51 (merged); reviews to be evidenced separately.

2 Vision

2.1 Preliminary Problem Statement

Business Background and Problem Description

Recreational players need bookable courts near their desired start time, while owners may want to offer lower prices to fill otherwise idle near-term slots. A late cancellation can reopen a usable slot, but the owner may lack an easy way to publish a targeted short-notice offer. Separately, an organizer who has already reserved a court but lacks enough players often recruits participants through scattered chat messages, making headcounts and last-minute withdrawals hard to track. These are product hypotheses from team discussion, not findings from a completed local user study.

The existing reservation flow also needs reliable court schedules and conflict prevention. Venue owners may manage courts across phone calls, messages, and spreadsheets, creating stale availability and conflicting booking requests. CourtMate therefore retains atomic booking validation while exploring two focused improvements: owner-controlled promotional prices for eligible near-term slots (Should F09) and helping organizers recruit participants for an already-booked session (Must F10). The team will validate whether these flows improve on current alternatives rather than claiming uniqueness based on feature names.

Proposed Solution Overview

CourtMate is a responsive web platform for court discovery, overlap-safe booking, cancellation, and owner management. Its new mandatory enhancement is a small Open Play Room linked to an already-confirmed booking. As an optional Should-have, Last-Minute Deals lets a Court Owner manually offer a lower price on an eligible unbooked slot starting within the next 24 hours. Players book through the existing reservation flow, subject to the same overlap checks.

- **For Players and Room Organizers:** Search courts, inspect available slots, reserve or cancel an eligible booking, create a public room for one's own confirmed booking, and join or leave an open room within its capacity. If F09 (Should) is implemented, browse owner-created Last-Minute Deals and book at a validated promotional price. Any per-person cost shown is indicative only; CourtMate does not collect deposits or payments.
- **For Court Owners:** Manage court details, operating hours, maintenance blocks, and incoming bookings. Availability, including reopened cancellations, is recalculated using authoritative booking data. If F09 (Should) is implemented, an owner may manually create or deactivate a discounted deal for an eligible unbooked slot; no automatic repricing is introduced.

The core journey is "discover available court $\rightarrow$ book safely $\rightarrow$ recruit participants through a booking-linked room." Engineering emphasis is placed on booking overlap protection, room ownership, capacity, and cancellation lifecycle. Optional deals are not required for the Must-have PA5 baseline.

Operating Environment

- **Client Environment:** The application operates as a responsive web interface accessible via modern desktop and mobile web browsers supporting HTML5, CSS3, and JavaScript (e.g., Google Chrome,

Mozilla Firefox, Microsoft Edge, and Apple Safari). No separate native mobile application (iOS/Android) is required.

- **Server and Cloud Environment:** The backend REST API and relational database are hosted on cloud infrastructure (planned deployment on Vercel for the frontend, Render for the backend service, and Neon for managed PostgreSQL), accessible over HTTPS.
- **User Characteristics:** Users include Players (including organizers and room participants) and Court Owners using modern browsers. Room organizer and participant are Player actions, not new system roles; server-side checks restrict the room to the creator's confirmed booking and prevent unauthorized edits.

Design and Implementation Constraints

- **Technology Stack Constraints:**
  - Frontend: ReactJS for booking, owner views, public room discovery, and basic room create/join/leave. Optional screens include Court Owner deal creation/deactivation and Player deal discovery (F09), plus editable account profiles (F08).
  - Backend: Python with FastAPI for REST booking/cancellation, room capacity and booking ownership validation. Optional F09 adds owner-authorized deal management and offer validity/price checks in the existing booking transaction; it is not a dependency of Must-have booking or rooms.
  - Database: PostgreSQL for courts, schedules, bookings, rooms, and room participants. Constraints and transactions protect reservation overlap and room capacity during concurrent join requests.
  - Testing and CI: Pytest / FastAPI TestClient with GitHub Actions. Must tests cover bookings, cancellation, room ownership, joins/leaves, maximum capacity, booking cancellation, and race conditions. If F09 is implemented, test deal ownership, slot eligibility and reopened cancellations, expiration/deactivation, stale prices, and concurrent booking conflicts.
- **Schedule and Resource Constraints:** The system is developed by a four-member student team (T05 – Nexora) following a strict Waterfall phase-gate schedule (PA1 through PA5), with core implementation and testing concentrated within Weeks 9–10, after test planning in Week 8.
- **Financial and Scope Constraints:** Online payments, deposits, automatic penalties, refund settlement, and automatically generated last-minute discounts are excluded. Manual owner-created F09 discounts are optional; no pricing engine is planned. Venue payment remains outside CourtMate (typically the organizer pays the venue), and any contribution from room participants is handled directly among people, without platform verification.
- **Process and Documentation Standards:** All deliverables must maintain strict traceability across PA baselines. Source code and documentation changes on the main branch require pull requests, peer reviews, and Conventional Commits formatting.

Assumptions and Preliminary Open Issues

PA2 must finalize fixed slot durations, booking/cancellation cutoff rules, organizer-only room editing, public room visibility, maximum participants, and the behavior of a room after booking cancellation. F09 (Should) needs agreement on its rolling 24-hour window, eligible slots, manual discount values, expiry/deactivation, and price confirmation; F08 profile editing also remains Should-have. For the Must MVP, only the owner

of an active confirmed booking can create a room, and room participation does not reserve an unbooked court.

2.2 Product Position Statement

CourtMate's Must-have positioning is a reliable end-to-end reservation with a lightweight, booking-linked room for finding enough players. This keeps community coordination traceable without building payments, matchmaking, automatic no-show fines, chat, or tournament management. Owner-managed Last-Minute Deals for eligible near-term slots are a non-blocking Should-have feature, without automatic pricing or payment processing. Similar commercial capabilities already exist; the team will measure whether the combined journey improves a specific task instead of claiming a market first.

Table 2.1: CourtMate product position statement

Element	Content
For	Sports players and sports court owners.
Who	Need to find near-term vacant courts and reliable bookings, and sometimes recruit enough people for a court already reserved; owners need fewer idle slots and dependable schedules.
The CourtMate	Is a responsive web platform for sports court booking, owner scheduling, and player coordination.
That	Lets players search and reserve available courts, then recruit other players through public rooms linked to confirmed bookings. It also lets owners manage courts and reservations.
Unlike	Fragmented booking through calls or chat and existing platforms that provide separate venue booking or social-play tools; their specific limitations require practical comparison.
Our product	Offers a lightweight booking-to-room workflow with server-side conflict prevention and simple roster management. Optional F09 lets Court Owners discount eligible slots within 24 hours and Players book at the promotional price verified by the server.

2.3 Stakeholders and Users

Table 2.2: CourtMate stakeholders and their main interests

#	Stakeholder / User	Description	Main Interests
1	Player	Players who discover/book courts; room organizers who already have a confirmed reservation; players who want to join a scheduled open session.	Find bookable courts and, if F09 is delivered, owner-offered discounts for slots starting within 24h; create/manage a booked-room roster, and join or withdraw within room rules.
2	Court Owner	End users (venue operators or staff) who list and manage sports courts on the platform.	Publish/update courts and operating hours, block maintenance slots, see booking statuses, and ensure eligible cancellation slots return to accurate availability; optionally create/deactivate owner-defined promotional prices for eligible unbooked slots (F09 Should).

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Table 2.2: CourtMate stakeholders and their main interests (Continued)

#	Stakeholder / User	Description	Main Interests
3	Nexora (Team T05)	The four-member student engineering team responsible for analyzing, designing, implementing, testing, and deploying CourtMate.	Delivering the agreed Must-have features on schedule, maintaining software quality and scope consistency across PAs, and completing the required deliverables.
4	Course Instructor & Teaching Assistants (TAs)	Academic evaluators and project supervisors from the Software Engineering Department (FIT – HCMUS).	Evaluating requirement completeness, traceability, architectural soundness, teamwork collaboration evidence, adherence to the Waterfall process, and working software delivery.

2.4 Scope and Key Features

The revised proposal has five core capability groups and ten numbered subfunctions: F01–F07 and F10 are Must-have; F08 Profile Management and F09 Last-Minute Deals are Should-have. F09 sits under the discovery group as an optional enhancement, not a sixth core group. F08 is no longer required for PA5; Legacy Ratings & Reviews and broad promotion/coupon campaigns are excluded; narrowly scoped, manually created F09 discounts are the sole proposed pricing exception.

Five Core Product Capabilities

The five groups below organize eight Must-have subfunctions, with F08 and F09 listed as optional enhancements to their relevant groups. The Must commitment includes one booking-linked Open Play Room flow (F10); advanced community and payment functionality remains out of scope.

Table 2.3: Five core features and traceability to detailed subfunctions

Core capability group	Included functions (Must/Should)
C1 Accounts and role-based access	F01 Authentication & Authorization (Must); F08 editable Profile (Should).
C2 Court discovery and optional owner deals	F02 Browse & Search (Must); F03 Details & Availability (Must); F09 Manual Last-Minute Deals (Should).
C3 Reservation and cancellation	F04 Court Booking (conflict prevention); F05 My Bookings and Cancellation.
C4 Court operations and owner oversight	F06 Court & Schedule Management; F07 Owner Booking Management.
C5 Open Play Room coordination	F10 booking-linked Open Play Rooms: create, view, join, leave within capacity (Must).

Table 2.4: MoSCoW subfunctions and implementation scope

ID	Feature	Description	Priority (M/S/C/W)
F01	Authentication & Authorization	Users can register, sign in, and sign out; the system enforces role-based access control separating Player and Court Owner permissions.	Must
F02	Browse & Search Courts	Players can browse listed active courts and filter or search by court name, sport type, and geographical area/location.	Must
F03	Court Details & Availability	Users can view detailed court information, including location, supported sport, facility descriptions, hourly price, and available dates and time slots.	Must
F04	Court Booking	Players can select an available date and time slot and submit a reservation; the server validates slot availability and uses database transactions/constraints to prevent conflicting or overlapping bookings.	Must
F05	My Bookings	Players can view their upcoming and past reservations and cancel an eligible booking in accordance with defined cancellation rules.	Must
F06	Court & Schedule Management	Court owners can add, edit, and deactivate courts they own, configure regular operating hours, and mark specific time slots as unavailable.	Must
F07	Owner Booking Management	Court owners can view and monitor reservations made for their own courts, including booking dates, time slots, player contact details, and booking status.	Must
F08	Profile Management	Authenticated users may edit non-critical profile information (name, contact preferences). Basic account identity, login, and role checks remain part of F01 Must.	Should
F09	Last-Minute Deals	Optional: a Court Owner manually creates or deactivates a discounted deal for an eligible, unbooked slot starting within the next 24 hours. Players browse offers and book through the normal reservation flow. The backend revalidates availability and offer validity and records the accepted promotional price. No automatic pricing or online payment.	Should
F10	Open Play Room Management	Must: the owner of an active confirmed booking may create one public room for that booking and set capacity; authenticated Players view/join/leave. Enforce booking ownership, unique membership, and capacity; close the room if the booking is cancelled. No payment, deposit, chat, or waitlist.	Must

Out of Scope (Won't-have for this Project)

To keep five core capabilities feasible during W09–W10, the following are explicitly outside the proposed MVP:

- **Payments and Financial Commitment:** No gateway, deposits, automatic no-show fees, chargebacks or

refunds, or escrow. Joining a room expresses intent, not a financially binding commitment; the organizer handles any venue payments outside the platform.

- **Automatic Pricing and Broad Promotions:** No rule-driven discounts, coupon campaigns, loyalty pricing, or automatic repricing. The only proposed exception is F09 (Should): a Court Owner manually creates/deactivates a lower-price deal for an eligible available slot starting within 24 hours, with validity checked at booking.
- **Native Mobile Applications:** No separate Android or iOS app; deliver a responsive web interface only.
- **Live Messaging and Push Alerts:** No room chat, SMS, email, or push gateway; roster and status updates are available when users refresh or revisit the web page.
- **Financial/POS and Multi-Branch Operations:** No staff shifts, inventory, loyalty/coupon engine, advanced analytics, or multi-branch admin.
- **Advanced Social and Scheduling:** No availability polls, teammate-matching algorithm, clubs, tournament brackets, member reputation, room waiting lists, automated penalties, or check-in. F08 profile editing and F09 owner-managed Last-Minute Deals remain optional Should-have only.

2.5 Representative Use Scenarios

The two scenarios demonstrate (1) reliable Must-have booking with an optional owner-created Last-Minute Deal, and (2) the Must-have room linked to a confirmed booking. They are proposed acceptance scenarios, not evidence of completed interviews or a market comparison.

Scenario 1 – Core booking with an optional owner-created deal. At 16:30 a Court Owner sees an available badminton slot at 19:00 and, if F09 (Should) is delivered, manually offers a discounted price for that slot. A Player browses near-term deals, sees the offer, and chooses to book. FastAPI checks slot availability and offer validity again at confirmation, stores the accepted price in the booking, and rejects an expired/deactivated deal or conflicting booking without silently changing the displayed price. Without F09, the Player books at the regular owner-set price. (Must F02–F05; optional F09.)

Scenario 2 – An organizer recruits players for a pre-booked court. A player has a confirmed booking for Saturday 19:00–21:00 and needs three participants. They open a public room linked to that booking, set a capacity, and share the room. Other authenticated Players join/leave; the roster remains within capacity despite concurrent requests. If the booking is cancelled, the room is closed. Costs, deposits, and penalties are handled outside CourtMate and are not verified by the app. (Must F01, F04–F05, F10.)

2.6 Competitive Landscape

Table 2.5: Public alternatives and scoped comparison (claims limited to published features)

Product	Publicly described capabilities	Comparison with CourtMate MVP
SanBong.vn	Sports venue discovery, listed availability, and online court booking; management tools for venues.	Booking/discovery overlaps. Test whether the Must booking-to-room journey reduces manual coordination; optional F09 may be evaluated through an owner-publishes-discount/player-books-offer journey.
ALOBO	Online booking plus operator tools such as booking calendars, customer management, revenue reporting, and add-on services.	Operator booking calendars overlap. CourtMate targets a narrower booking-linked room workflow without POS/payment; no unverified superiority claim.
ONTO	Sports venue discovery, booking, and player/community-oriented features; an owner-facing offering is also described.	Community features overlap with the proposed Room capability. Verify the same "recruit players for a booked court" task before claiming differentiation.

Sources for PA2 competitor task checks: https://sanbong.vn/, https://www.alobo.vn/, and https://onto.vn/. PA1 does not claim verified local interviews or that these products lack booking-linked rooms or manual-deal functions. Compare a "recruit three people for an already-booked court" journey and, if F09 is attempted, an owner-publishes-discount/player-books-offer journey.

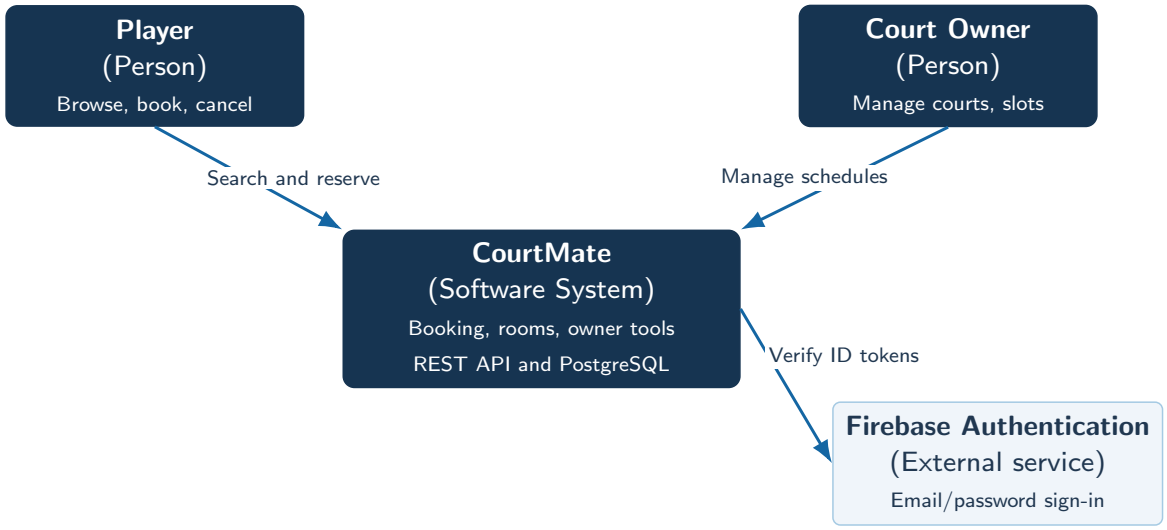
3 Proposed Solution

3.1 Preliminary Architecture

CourtMate continues with two application roles (Player and Court Owner). The core booking and owner workflows remain intact. F10 adds public Open Play Rooms tied to exactly one active confirmed booking of the Player organizer, with room membership and participant-capacity checks. Optional F09 supports owner-authorized creation/deactivation of deals for available slots beginning within the next 24 hours and Player discovery of these deals. It reuses the court/booking data, with server-side offer validity and promotional-price checks before booking confirmation. The System Context figure retains the same actors and external services; PA3 will detail the Room and Deal components.

Firebase Authentication is the planned email/password identity provider. FastAPI verifies ID tokens and uses PostgreSQL authorization data to enforce booking ownership and room-organizer rights. Room membership and booking status must be checked server-side; joins cannot exceed capacity, and cancelling the source booking closes the room. No additional role, chat, deposit, matchmaking, push system, or payment gateway is introduced. F08/F09 remain optional and do not block the Must demo.

Figure 3.1: CourtMate preliminary C4 System Context (scope-aligned)



System boundary: CourtMate frontend, API and database

3.2 Technology Stack

The stack remains React/Vite, FastAPI, Firebase Authentication, PostgreSQL (Neon), Vercel, Render, Pytest, and GitHub Actions. Must functionality adds room APIs and two tables (rooms and memberships), not new infrastructure. Optional F09 requires a deal model/API, owner controls and checkout-time price validation, but no automatic pricing engine or new infrastructure. It will be deferred if core functionality or testing is behind schedule. Backend and database, rather than client state, enforce booking and room concurrency rules.

Table 3.1: Technology stack and responsibilities

Layer	Technology	Purpose / design choice
Frontend	ReactJS + Vite	Must: booking/player/owner and Room screens. Should: owner deal create/deactivate and Player deal discovery screens (F09), and editable profile (F08).
Backend	Python + FastAPI	Booking/cancellation, room creation/authorization/join/leave/capacity APIs; optional F09 deal management and booking-time offer/price verification.
Database	PostgreSQL (Neon)	Users/roles, courts, bookings, rooms, and room membership. Database transactions/constraints for booking and room consistency.
Authentication	Firebase Authentication (planned)	Email/password sign-in; room participants and organizers are Players. FastAPI verifies Firebase tokens.
Authorization	FastAPI + PostgreSQL	Owner ID, organizer ID, and booking ownership checks; room membership authorization and lifecycle validation.
Hosting / deployment	Vercel + Render + Neon	Frontend, backend API, and managed PostgreSQL, respectively.
Testing & CI	Pytest, FastAPI TestClient + GitHub Actions	Must tests for booking/room edge cases and CI; F09 tests for offer ownership, expiry/deactivation, price confirmation and booking conflicts if implemented.

3.3 Hardware and Operating Environment

Development Hardware

- Four existing student laptops/desktops; dual-core processor or better and at least 8 GB RAM (16 GB recommended).
- Approximately 10 GB of free storage for code and local dependencies; stable internet access for Git and cloud services.

Software and Development Environment

- Windows, macOS, or Linux; Visual Studio Code, Git, Node.js/npm for React, and Python/venv for FastAPI.
- Managed Neon PostgreSQL for persistent application data; local database testing can be performed separately.
- Pytest and FastAPI TestClient for unit/API tests, with GitHub Actions for continuous integration.
- Modern browsers (Chrome, Edge, Firefox, Safari) for responsive web testing.

Deployment and Operating Constraints

The React application is hosted on Vercel, the FastAPI API service on Render, and PostgreSQL on Neon. The React frontend calls FastAPI over HTTPS; only the backend uses the database connection string.

Production credentials belong in platform environment variables, not in source control. Backend CORS configuration must allow the deployed frontend origin. Free-tier quotas and cold starts must be validated before the live demo.

Core availability is always read from current booking/blocked-slot data; a search result is not itself a reservation guarantee. Confirming a booking requires database-level overlap safety. The room owner must own an active confirmed booking, and a unique membership/capacity check protects concurrent joins; cancelling the booking closes its room. Room-join cutoff rules are finalized in PA2. If optional F09 is delivered, an owner can create a time-limited lower-price offer only for an eligible unbooked slot within the rolling 24-hour window. Backend booking checks offer status, ownership, expiry and the displayed promotional price, rejects stale offers and saves the accepted price with the booking. No deposit or online payment is planned.

4 Project Plan

4.1 Process and Phase Gates

CourtMate follows the course's Waterfall phase-gate process. Each Project Assignment (PA) produces a reviewed deliverable that becomes the baseline for the next phase; any approved changes are recorded in a later PA. Teaching-staff approval of PA0 is a prerequisite under the course guidelines. Official Moodle timestamps govern the submission deadlines.

Table 4.1: Project phase gates and exit criteria

PA	Phase	Due	Exit criteria
PA1	Initiation & planning	End of Week 3	Reviewed vision, C4 Context, WBS, effort, Gantt, roles, risk register, and collaboration setup.
PA2	Requirements	End of Week 5	Numbered functional/nonfunctional requirements, use cases, mockups, and traceability reviewed.
PA3	Design	End of Week 7	Reviewed C4/ADRs, database ERD, UML diagrams, and UI/API specifications.
PA4	Test planning	End of Week 8	Test plan with EP, BVA, decision tables, repeatable test cases, and requirement coverage.
PA5	Implementation, testing & deployment	End of Week 10	Must-have system deployed; tests, defects, coverage, CI, automated scenarios, and demo evidence.
Final	Presentation & demo	Week 11	Live demo, presentation, individual Q&A, and final handoff.

4.2 Work Breakdown Structure (WBS)

The WBS retains three levels (phase → workstream → single-owner work package). It now has 41 Level-3 packages: the original 38, plus one optional Last-Minute package and two separate Must-have Room packages (backend and UI). The original Authentication/Profile and Login/Profile packages are slightly reduced to a basic account flow, because editable F08 profiles are deferred. Owners and package IDs follow the supporting WBS/Gantt/CPM Sheet, The WBS and effort estimates are consistent with the linked workbook. Dates and resource capacity remain provisional and require team confirmation.

Level-2 workstreams: 1.1 Vision; 1.2 Solution; 1.3 Planning; 2.1 Analysis; 2.2 Validation; 3.1 Architecture; 3.2 Detailed Design; 4.1 Test Strategy; 4.2 Test Cases; 5.1 Core Platform; 5.2 Booking, Rooms & Optional Deals; 6.1 Presentation; 7.1 Management. New PA5 packages are 5.2.8 (F09 Should), 5.2.9/5.2.10 (F10 Must).

The 41 work packages each have one accountable owner and an effort estimate. This document proposes increasing package 5.2.8 from 8 h to 18 h for owner-managed Last-Minute Deals (F09 Should), resulting in 302 h for all proposed features while retaining 108 Must hours in PA5. The linked Sheet reflects the revised 18 h estimate and 302 h total. F09 remains outside the Must critical path;

Supporting workbook: *T05-CourtMate – WBS, Gantt, CPM & Cost*.

4.3 Estimation

Expert judgment estimates the total at 302 person-hours: 271 h original + 18 h optional F09 + 8 h F10 Room API + 7 h F10 Room UI − 2 h from deferring editable F08 profile work. PA5 is 126 h, of which 108 h are Must and 18 h are optional (F09 Should). The minimum planned Must-only project remains 284 h (108 h PA5); extra profile editing is not included in these totals. With all planned features PA5 averages 15.75 h/member/week (13.5 for Must-only) over W09–W10, above PA0's typical 5–7 h; capacity and prioritization need confirmation. AI-assisted coding is not a substitute for tests.

Table 4.2: Level-3 work packages and effort estimates (41 packages; includes optional F09)

WBS ID	Work package	Owner	Effort (h)	Method / assumptions
1.1.1	Problem & product position statement	Quang	6	Expert judgement
1.1.2	Stakeholders, MoSCoW & exclusions	Quang	6	Expert judgement
1.2.1	C4 System Context diagram	Hiếu	4	Expert judgement
1.2.2	Technology stack & operating environment	Hiếu	4	Expert judgement
1.3.1	WBS, estimates, Gantt, CPM, roles & cost	Hưng	9	Expert judgement
1.3.2	Risk register & mitigation	Trung	5	Expert judgement
1.3.3	Tools, collaboration & PA1 integration	Hưng	7	Expert judgement
2.1.1	Actors, roles & booking business rules	Quang	6	Expert judgement
2.1.2	Functional / nonfunctional requirements & AC	Quang	9	Expert judgement
2.1.3	Use-case model and specifications	Trung	8	Expert judgement
2.2.1	Main UI mockups and user journeys	Hiếu	8	Analogy / UI
2.2.2	Traceability, review & PA2 submission	Hưng	7	Expert judgement
3.1.1	C4 Container / deployment & ADRs	Hiếu	9	Expert judgement
3.1.2	Database ERD, constraints & indexes	Quang	9	Expert judgement
3.2.1	Class, sequence & state diagrams	Hiếu	10	Expert judgement
3.2.2	UI specifications & API contracts	Hưng	8	Analogy / UI
3.2.3	Consistency review & PA3 submission	Trung	6	Expert judgement
4.1.1	Test plan, levels & environment	Trung	5	Expert judgement
4.1.2	EP, BVA & decision tables	Quang	6	Expert judgement
4.2.1	Repeatable test cases & requirement coverage	Hiếu	9	Expert judgement

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Table 4.2: Level-3 work packages and effort estimates (41 packages; includes optional F09) (Continued)

WBS ID	Work package	Owner	Effort (h)	Method / assumptions
4.2.2	PA4 peer review & submission	Trung	3	Expert judgement
5.1.1	Backend/frontend scaffold, database & CI	Hưng	6	Expert judgement
5.1.2	Authentication, basic account display & tests (F01 Must; F08 edit Should)	Hưng	7	Expert judgment; editable profile deferred
5.1.3	Court, schedule & availability APIs	Quang	10	Expert judgement
5.1.4	Browse/search/detail screens	Hiếu	7	Analogy / UI
5.1.5	Login and basic account display screens (F01 Must; F08 edit Should)	Hiếu	4	Analogy/UI; editable profile deferred
5.2.1	Atomic booking API & concurrency tests	Hưng	11	Analogy + risk buffer
5.2.2	Cancellation & owner booking APIs	Quang	8	Expert judgement
5.2.3	Booking/My Bookings/owner screens	Hiếu	11	Analogy / UI
5.2.4	Unit coverage, E2E automation, regression	Trung	12	Expert judgement
5.2.5	Staging, production deploy & CI verification	Quang	8	Expert judgement
5.2.6	Regression results & defect report	Trung	5	Expert judgement
5.2.7	Final smoke check, demo video & PA5 evidence	Trung	4	Expert judgement
5.2.8	Optional owner-managed Last-Minute Deals: create/deactivate, Player discovery, price validation, UI & tests (F09 Should)	Quang	18	Proposed revision from 8 h; manual discounts and checkout validation;
5.2.9	Room API: ownership, join/leave, capacity tests (F10 Must)	Hưng	8	Analogy; minimal booking-linked room
5.2.10	Open Play Room UI: list/detail/join and integration (F10 Must)	Hiếu	7	Analogy/UI; no chat, payments, or waitlist
6.1.1	Presentation slides & demo storyline	Hiếu	6	Expert judgement
6.1.2	Team rehearsal & demonstration checks	Hưng	5	Expert judgement
6.1.3	Retrospective, Q&A & final delivery	Trung	5	Expert judgement
7.1.1	W02–W10 weekly reports & meetings	Hưng	9	Expert judgement
7.1.2	Board updates, decisions & risk follow-up	Quang	7	Expert judgement

Estimated effort by phase (PA5 includes 108 Must + 18 optional hours; proposed F09 revision):

Table 4.3: Effort allocation by project phase (302 person-hours including optional F09)

Phase	Effort (person-hours)	Planned weeks
PA1 – Initiation & planning	41	W02–W03
PA2 – Requirements	38	W04–W05
PA3 – Design	42	W06–W07
PA4 – Test planning	23	W08
PA5 – Implementation/testing	126	W09–W10
Final – Presentation	16	W11
Ongoing – Weekly management	16	W02–W10
TOTAL	**302**	

Planned owner effort: Hưng 77 h, Quang 93 h, Hiếu 79 h, and Trung 53 h (302 h total). These are estimated workloads, not measured contribution shares.

PA5 capacity risk: 126 h over W09–W10 (Hưng 32 h, Quang 44 h, Hiếu 29 h, Trung 21 h), above PA0's 5–7 h/person/week target. Implement 108 Must hours first; optional F09 adds 18 h only if capacity permits. Confirm availability and raise any scope changes with the TA.

4.4 Schedule

This document proposes 302 h across 41 packages; In W09–W10, Must Room API (5.2.9; 03–04 Dec) precedes Must Room UI (5.2.10; 05 Dec), and both feed the 06 Dec final smoke/demo gate. Optional F09 owner-managed Deals (5.2.8; previously scheduled 30 Nov–05 Dec) remains off the Must path, but its 18 h proposal requires resource/duration re-estimation to confirm schedule feasibility. The 14-day CPM is an illustrative booking/room subnetwork; fixed course phase gates and resource loading remain separate constraints.

Phase windows follow the course calendar: PA1 W02–W03, PA2 W04–W05, PA3 W06–W07, PA4 W08, PA5 W09–W10, and Final W11. W02 is taken as 05–11 October 2026 for the proposed weekly Gantt; exact submission cutoffs follow Moodle.

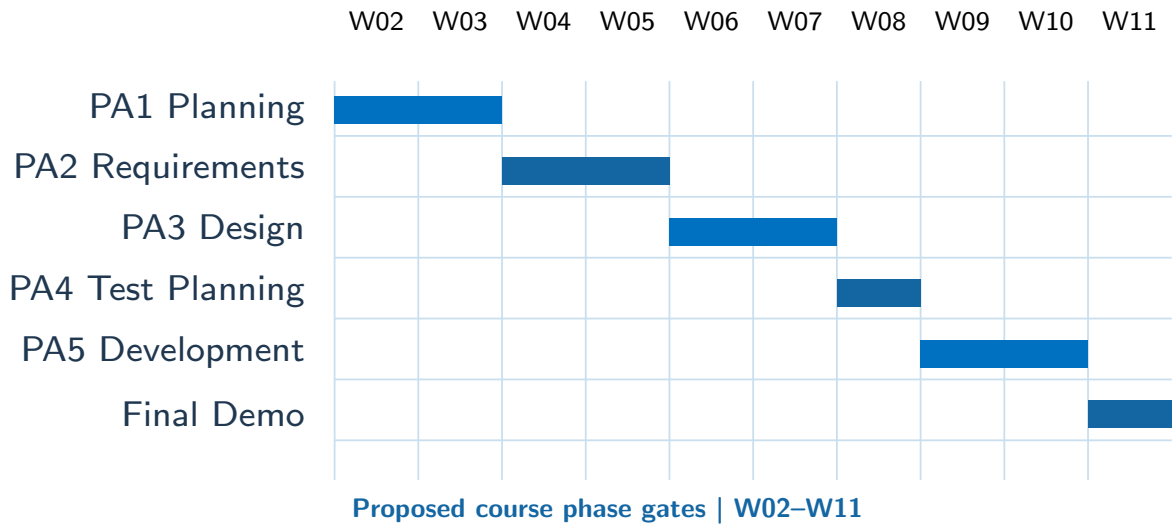
Detailed schedule and task dependencies: Open WBS/Gantt workbook.

Key dependencies: PA0 approval → PA1 proposal → PA2 requirements → PA3 design → PA4 test planning → PA5 build/test/deployment → Final demo. Tasks inside a phase may overlap when dependencies and member availability allow.

Critical path (phase-gate level): PA1 (2 weeks) → PA2 (2) → PA3 (2) → PA4 (1) → PA5 (2) → Final (1) = 10 calendar weeks (W02–W11) with zero slack under the fixed phase-gate assumptions.

Illustrative PA5 CPM: A→B→D→E→F and A→B→D→G→H→F are two 14-day critical branches. G is the Room API package (5.2.9, 2 days) and H is Room UI (5.2.10, 1 day), both after core booking; optional F09 is outside the Must critical path. See the linked CPM & Cost Sheet for provisional ES/EF/LS/LF; the optional F09 package dates require reassessment against team capacity. Resource feasibility remains

Figure 4.1: CourtMate phase Gantt (weekly timeline, proposed)



provisional. Scaffold A (2 days) → Court/availability API B (5) → Atomic booking API D (3) → Booking UI integration E (3) → Release/demo gate F (1) = 14 elapsed days, from 23 November to 06 December. Browse/detail UI C (5 days) runs in parallel after A with 3 days of float. This is not the verified whole-project critical path.

4.5 Roles and Responsibilities

Table 4.4: Project roles and responsibilities

Role	Main	Backup	Responsibilities
Team lead / Project manager	Hưng	Quang	Scope, assignments, schedule, risk escalation, weekly reports, and TA contact.
Business analyst	Quang	Trung	Elicit rules and requirements; organize review and traceability.
Designer / Architect	Hiếu	Quang	C4/UML, UI design, API, and database architecture coordination.
Developers	Hưng, Quang, Hiếu	Trung	Implement and unit-test modules; review PRs and integrate functionality.
Software tester	Trung	Hiếu (proposed)	Test plan, EP/BVA, test cases, automation, and defect/test reports.

PA1 review assignments: Quang prepared Section 2 (Hưng reviews); Hiếu prepared Section 3 (Trung reviews); Hưng prepared Sections 4 and 6 (Quang reviews); and Trung prepared Section 5 (Hiếu reviews). Authorship, PR creation, and approvals are recorded separately. Role backups for later phases follow the PA0 team contract unless a documented team decision changes them.

4.6 Human Resources and Cost Estimate

The four-person team retains React/Vite, FastAPI, Firebase Authentication, Neon PostgreSQL, Vercel, and Render. The plan now includes two Must Room work packages (8+7 h) and an optional Should Last-Minute Deals package (18 h), while reducing the original account/profile packages by 2 h in total. Effort represents estimated labor value, not cash wages.

Table 4.5: Cost estimation and assumptions

Item	Calculation / assumption	VND
Equivalent labor value	302 h × 80,000 VND/h	24,160,000
Contingency	10% × labor value	2,416,000
Optional domain	Illustrative allowance	250,000
Hosting/database	Provisional free-tier budget	0
TOTAL (illustrative)	**Including optional domain**	**26,826,000**

The proposed revised plan estimates 302 person-hours with an illustrative labor rate of 80,000 VND/h, 10% contingency, and an optional 250,000 VND domain. The resulting economic total is 26,826,000 VND (24,160,000 VND labor + 2,416,000 VND contingency + 250,000 VND domain). Direct cash expense remains 0 VND if free tiers suffice and the domain is not purchased; extra hosting charges are excluded.

The linked Sheet now reflects the revised 302-hour plan across 41 work packages, including the updated owner allocation and cost estimates. Dates, dependencies, Gantt, and illustrative CPM remain planning assumptions subject to resource feasibility review. Actual hours, delivery results, and documented changes must be tracked separately from proposed estimates.

5 Risk Management

The register covers the original ten risks and two scoped additions: R11 applies only if the optional F09 owner-managed Last-Minute Deals feature is implemented; R12 covers room vacancies, cancellation, and capacity overflow in the mandatory F10 workflow. Risk Exposure (RE) is assessed using the lecture's five-level scales, rather than a 1–10 scale: RE = Probability × Loss. Probability scores are Unlikely (10), Seldom (25), Occasional (50), Likely (75), and Frequently (90). Loss scores are Negligible (10), Insignificant (25), Moderate (50), Serious (75), and Catastrophic (90). These scores are provisional judgments for prioritization, not measured incident frequencies or actual monetary losses. Owners remain proposed leads until the team confirms assignments.

Table 5.1: Risk register and mitigations

ID	Risk	Type	P	L	RE	Mitigation / contingency	Owner
R1	A member leaves the project	People	25	90	2,250	Cross-train a backup, hand over current deliverables and issues; reassign critical tasks. Request TA approval for any necessary scope/schedule change.	Hưng
R2	A member misses a committed deadline	People	50	50	2,500	Use Issues with owners and due dates; surface blockers within 24h, agree on a recovery plan, and escalate/reassign before the next phase gate.	Hưng
R3	Noncompliance with commit and file-naming rules	Organizational	25	25	625	Keep naming rules in `docs/README.md` and `CONTRIBUTING.md`; check file paths and commit messages during PR review and in lightweight repository checks.	Hưng
R4	Excessive AI reliance without understanding	People	50	75	3,750	Require the author to explain deliverables, verify AI output with course material and peers, and disclose significant AI use in the PA log.	Hưng
R5	Hosting outage, free-tier limits, or configuration failure	Technology	25	90	2,250	Deploy a minimal working app early, check environment variables, health endpoint, and quotas; rehearse demo and prepare an offline walkthrough.	Quang
R6	Underestimated tasks / infeasible milestones	Estimation	25	50	1,250	Review remaining effort weekly against team capacity; reprioritize and negotiate revised allocations; escalate Must-have scope changes to TA.	Hưng
R7	Requirements do not match user needs or agreed scope	Requirements	25	90	2,250	Review feature scope with team/TA, define acceptance criteria in PA2, and trace changes across requirements, design, and tests.	Quang

Continued on next page

Table 5.1: Risk register and mitigations (Continued)

ID	Risk	Type	P	L	RE	Mitigation / contingency	Owner
R8	Frontend–backend integration incompatibility	Technology	75	50	3,750	Agree on API contracts and sample payloads in PA3; integrate small flows early and add repeatable contract/integration tests.	Quang
R9	Race condition causes double bookings	Technology	75	75	5,625	Use atomic PostgreSQL transactions and slot constraints/locking; add concurrent booking tests; reject or roll back conflicting requests.	Hưng
R10	Insufficient test coverage causes defective delivery	Quality / Process	75	50	3,750	Create PA4 test cases for Must-have flows, automate core API/booking scenarios in CI, log defects, and run regression plus pre-demo smoke tests.	Trung
R11	Expired, invalid, or incorrectly priced Last-Minute Deal	Technology	50	75	3,750	Authorize owner changes, validate available slot/24-hour eligibility, recheck offer status and quoted price within the booking transaction; reject stale offers and test concurrent claims.	Quang
R12	Room is underfilled, cancelled, or over capacity	Requirements / UX	75	50	3,750	Only create room for confirmed organizer booking; enforce capacity atomically, update roster/cancel state, show that joining is not a financial guarantee; test withdrawals and booking cancellation.	Hưng

Risk priority is led by R9 (RE 5,625: double bookings). R4, R8, R10, and R12 (RE 3,750 each) require close attention to AI-output verification, integration, testing, and room capacity/withdrawal. R11 (RE 3,750: invalid deals or stale booking prices) becomes an active risk only if optional F09 is approved for implementation. Tied scores indicate the same priority tier rather than an exact ordering. F10 does not guarantee attendance: joining a room is not a paid commitment; no deposit, penalty, or refund is collected.

Risk monitoring will be part of the weekly team status meeting. Each risk owner reviews probability, loss, current mitigation, and any warning indicators. The team will track failed concurrent-booking tests (R9), unresolved API contract/integration blockers (R8), uncovered or failing Must-have test cases (R10), and rejected/over-capacity room joins or booking cancellations (R12). If F09 is implemented, failed deal-expiry or price-consistency checks will also be monitored (R11). High-priority risks, new incidents, and changes in RE will be recorded in the weekly status report and escalated to the team lead/TA where scope or schedule is affected.

6 Tools & Collaboration Setup

The team uses GitHub for source control, Issues and PRs for change tracking, and Google Drive for collaborative documentation and planning. The course also requires a working task board, an accessible repository, protected main, independent PR review, and a communication channel with the TA. Table 6.1 separates evidenced activities from platform controls that still need direct verification.

6.1 Tool Setup and Evidence

Table 6.1: Collaboration tools, purposes, and verification status

Tool / platform	Purpose	Current state and verification
Moodle	Course announcements and PA submissions	Required submission channel. PA1 must be uploaded as PDF and editable source; no completed submission is claimed.
Messenger / Google Meet	Daily updates and online team meetings	Messenger/Meet used for routine coordination under PA0.
GitHub repository	Version control, team documentation, and contribution evidence	CourtMate repository, team documentation, and naming rules are available. Instructor/TA access and main-branch protection are required and must be evidenced.
GitHub Issues / Projects	Assign and monitor work packages	Issues #45–#49 cover the five PA1 section assignments. The revised WBS has 41 Level-3 work packages, not 41 verified live GitHub Project tasks. Each active task still needs evidence of assignee, deadline, and board status.
Pull requests / main controls	Peer review, integration, and code safety	PRs #52 and #53 were merged on 17 October; #54, and #55 were merged on 18 October; #51 were merged on 19 October. Confirm protected-main and independent-approval.
Google Docs / Drive	Shared draft editing, WBS/Gantt planning, meeting notes, and weekly-report references	CourtMate – T05 Nexora team folder and WBS/Gantt workbook are available on Google Drive. Use tracked changes/version history and actual review comments as evidence.

Evidence and shared workspace: CourtMate repository; PA1 Issues; Team Google Drive folder; WBS/Gantt workbook.

6.2 Task Tracking and Peer Review

The synchronized WBS/Gantt proposes 41 Level-3 work packages, including three new PA5 tasks (5.2.8 optional F09; 5.2.9/5.2.10 Must F10). PA1 Issues #45–#49 document the five section assignments; they are not the same as 41 implemented GitHub cards. The team should create or update feature work items as they start, with real owners, due dates, acceptance tests, and peer review, and should not claim that unverified board or protection settings are already complete.

Table 6.2: Current PA1 section assignments

Issue	PA1 section	Assigned to	Reviewer
#45	2 – Vision	Quang	Hưng
#46	3 – Proposed Solution	Hiếu	Trung
#47	4 – Project Plan	Hưng	Quang
#48	5 – Risk Management	Trung	Hiếu
#49	6 – Tools & Collaboration	Hưng	Quang

Review workflow. Issue → task branch → focused commit → Pull Request to main → independent peer review → merge after checks → board/status report update. Mark a task Done only when its reviewed deliverable and evidence are available.

Naming and security. Task branches use the type/short-task-name convention (for example, `docs/pa1-tools-collaboration`). Commits follow Conventional Commits (for example, `docs: add PA1 tools and collaboration section`). Passwords, API keys, and `.env` files are never committed. Independent PR review and protected-main settings remain explicit course requirements.

6.3 Communication and Weekly Reporting

Messenger is the team's daily coordination channel, while Google Meet supports remote discussion. PA0 schedules internal meetings for Tuesdays and Saturdays at 20:00 (UTC+7); key decisions and blockers are recorded so absent members can follow up.

Shared project workspace: *CourtMate – T05 Nexora* (Google Drive).

A 15-minute weekly status meeting and one report per week are required for W02–W10. The Team Lead (or backup) consolidates completed work, Issue/PR references, blockers, risks, and next actions. Reports are committed to `docs/management/weekly/Wxx-status.docx` (or `.md`); they are not submitted separately to Moodle.

Weekly reporting deadline. Monday at 22:00 (UTC+7) for W02–W10. The team must document its chosen cutoff and use it consistently, without rewriting previously committed reports.

6.4 Repository Structure and Submission

Table 6.3: Intended repository document structure

Folder / path	Purpose
`src/`	Frontend and backend implementation, added during PA5.
`docs/requirements/`	Vision, requirements, use cases, and mockups.
`docs/design/`	Architecture, C4/UML diagrams, ADRs, and UI design.
`docs/management/weekly/`	W02–W10 reports; other management materials remain under `docs/management/`.
`docs/test/`	Test strategy, test cases, test results, and defect reports.
`docs/ai-usage/`	Supporting AI records; does not replace each PA's mandatory AI Usage Log.
`pa/PAx/`	Reviewed final PAx (1 to 5) PDF and editable source, consistent with Moodle submission.

7 AI Usage Log

This log summarizes the AI assistance declared by CourtMate T05 members during PA1, based on their individual AI Usage Logs stored in the GitHub repository. AI tools were used to support documentation refinement, project planning, technical research, and architecture descriptions. All AI-generated suggestions were subject to human review and adjustment. The team remains responsible for final project decisions, estimations, and deliverables. No AI usage was reported by Le Thanh Trung.

#	Member	Tool	Task / purpose	How output was verified / changed
1	Nguyễn Đức Hưng	ChatGPT	Assisted with improving the problem statement and project objectives, organizing feature priorities using MoSCoW, reviewing effort estimates, and identifying potential project risks.	Reviewed and refined the wording, verified alignment with project goals, evaluated estimates against team capacity, and selected relevant risks and mitigation strategies.
2	Bùi Phạm Quang	ChatGPT / GitHub Copilot	Supported technology stack comparison, high-level system architecture planning, non-functional requirements, Git workflow recommendations, and technical feasibility assessment.	Selected technologies based on team familiarity and requirements, simplified the architecture, refined testable metrics, aligned repository structure with PA1, and adjusted the scope based on feasibility.
3	Dương Trung Hiếu	ChatGPT	Assisted with writing architecture descriptions and Hardware and Operating Environment content for the PA1 proposal.	Reviewed the suggested content using personal technical knowledge and feedback from other team members before incorporating it into the document.
4	Lê Thành Trung	None	No AI tools were used during PA1.	Not applicable.